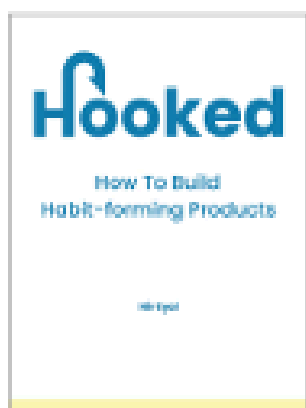




HOOKED

How to Build Habit-Forming Products

NIR EYAL



The Big "So What"

Why are certain products or services so addictive while others fail? In this book, Eyal explains the role of habits in successful products/services, and how you can [use the 4-step Hook Model to shape customer behaviors and habits](#).



Introduction



To have a massively successful business like Google or Facebook, it's not enough to have the marketing muscle to reach lots of people. You must learn to leverage human habits and emotions and get people to use your product so regularly that it becomes a habit.



Habit-forming companies offer first-to-mind solutions, i.e. they become so strongly associated with a problem/need that people automatically reach for these solutions when the triggers appear. For example, you'll automatically think of Google when you need to search for information, or start scrolling through your Facebook or Instagram feeds when you're bored or lonely. The question is, how do you develop such an association?



The key is to create a series of hooks and experiences that lock people into using your products habitually. After years of research, consulting and real-life experience, [Eyal developed the 4-part Hook Model which can be used to embed habit-forming cycles into a product/service](#). In this summary, we present the key ideas in 3 parts:

- Habits and habit-forming products

KEY QUOTES

"In order to win the loyalty of their users and create a product that's regularly used, companies must learn not only what compels users to click, but also what makes them tick."

"Hooks connect the user's problem with a company's solution frequently enough to form a habit."

- The 4 Steps of the Hook Model; and
- Applying the Hook Model

We'll use the term "products" to refer generically to all types of products, services, technologies and solutions.



Habits and Habit-Forming Products

Whenever we do something, it creates a neural pathway in the brain. The more we repeat the action/routine, the more it reinforces the pathway, until the action is triggered automatically by situational cues. For instance, we automatically go to the bathroom and brush our teeth once we wake up.



Benefits of habit-forming products/services

User habits can significantly increase a business' return on investment due to a combination of 4 key benefits.



Higher Customer Lifetime Value (CLTV). CLTV is the amount of money a customer spends with you before they stop using your products or die. When a customer uses your product frequently and habitually, CLTV increases.



Greater price flexibility. The more people use your product, the more they're willing to pay and the less price-sensitive they become. The success of free-to-play video games hinges on their ability to get users hooked to the game so they'll end up buying extra lives, tools or powers to move up the game levels. Likewise, many free apps/tools like Evernote rely on users' willingness to pay for upgraded features like collaboration tools or offline access.



Accelerated growth. The more people use and value a product, the more likely they'll share it with their friends. Viral Cycle Time (VCT) is the amount of time a user takes to invite another user. When people use your products several times daily, they initiate more loops (e.g. tagging

KEY QUOTES

"The aim is to influence customers to use your product on their own, again and again, without relying on overt calls-to-action such as ads or promotions."

"Hooked users become brand evangelists — megaphones for your company, bringing in new users at little or no cost."

Facebook friends several times daily) which generate more responses from other users to improve the VCT.



Competitive advantage. Once customers form habits around your products, it creates a real competitive advantage.

- Habits are so hard to change that a new solution must be $\geq 9\times$ better to break the inertia to change. The QWERTY keyboard was actually invented in the 1870s for typewriters. Commonly-used characters were spaced out to prevent the typewriter keys from jamming when pressed together. Today, we're still using QWERTY although typewriters are long obsolete. Better keyboards have been designed for modern computers, but the benefits simply weren't enough to motivate change.
- Even when you stop a habit and start a new one, it's easy to revert back into old habits since the neural pathways still exist and are easily reactivated. That's why two-thirds of alcoholics return to drinking within 1 year of finishing their rehabilitation programs. The only way to lock in a new habit is to keep doing it repeatedly for a long period of time.
- We also resist change if we've stored valuable data in the products, e.g. we don't want to lose our photos/memories on Instagram nor our emails in Gmail. These technologies also use customer data and algorithms to create even more hooks.



A solution can be a painkiller and/or a vitamin.

- Generally, people are ready to pay for painkillers which offer immediate relief from an existing pain (e.g. how Tylenol eases aches and pains). On the other hand, vitamins don't solve an obvious pain-point—they appeal more to emotions (e.g. the desire to be attractive or healthy) and require habit formation.
- Habit-forming products are both painkillers and vitamins. They start out as nice-to-have options. But once you become reliant on them, they become painkillers since you'll feel an "itch" or discomfort if you don't use them.

KEY QUOTES

"Old habits die hard and new products or services need to offer dramatic improvements to shake users out of old routines."

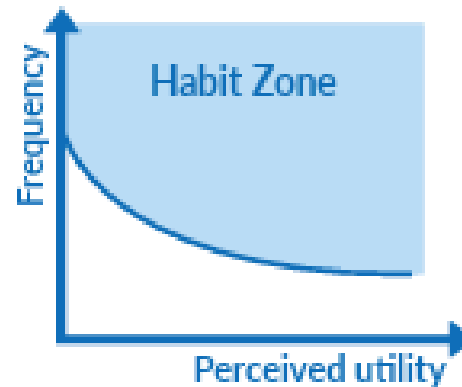
"A habit is when not doing an action causes a bit of pain."

"Once we come to depend on a tool, nothing else will do."



A product's habit-formation potential hinges on 2 factors:

- Frequency of use: how often the behavior occurs; and
- Perceived utility: how useful or rewarding it is compared to other alternatives.



A behavior falls into the "Habit Zone" only when it happens frequently enough and has a high-enough perceived utility. By embedding the 4-step process of the Hook Model into your products, you can move your product into this zone.



ACTION STEPS

Identify the habits required for your business model. Ask yourself:

- What problem does your product solve for your users?
- How else do users currently solve that problem? Why do they need your product?
- How frequently are users likely to use your product?
- What user behavior should you make into a habit?

KEY QUOTES

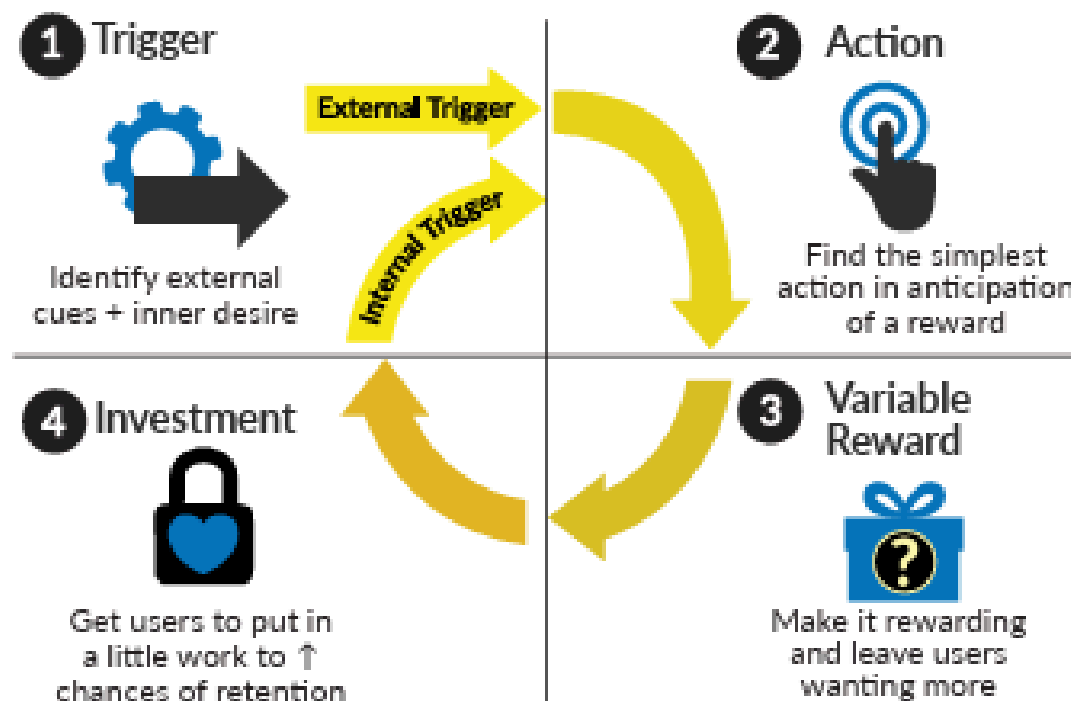


The 4 Steps of the Hook Model



The Hook Model involves 4 steps that run in a loop or cycle: Trigger → Action → Variable reward → Investment.

The Hook Model





Step 1: Trigger



Habits happen when a trigger automatically brings about a behavior. Such habits are formed in layers over time; they aren't developed overnight.



There are 2 types of triggers:

- External triggers provide an obvious call-to-action, i.e. they tell people what to do. These can come in the form of pictures, text, and/or explicit instructions e.g. "Click here" or "Call this number". A well-designed trigger has so much information embedded that it doesn't require long instructions on what to do. There're 4 key types of external triggers:
 - (i) Paid triggers (e.g. search engine marketing, advertisement) which can be costly in the long-run.
 - (ii) Earned triggers (e.g. media coverage, viral videos, features in app stores) which can bring temporary spikes in numbers.
 - (iii) Relationship triggers (e.g. word-of-mouth, electronic invitations, social media "likes").
 - (iv) Owned triggers come from having a space in the user's environment (e.g. an app icon on their phone, newsletters in their inbox).

Generally, (i)-(iii) are good for acquiring new users, while (iv) helps to trigger repeated behavior until a habit is formed.

- Internal triggers. An internal trigger is formed when a product becomes tied with a thought, routine, or emotion. When a product relieves a pain-point, you associate certain feelings with it. As the experience is repeated, the association is strengthened, until you automatically reach for those products for quick relief whenever you experience the internal triggers. That's why smokers feel the urge for a cigarette when they're stressed. People turn to Google when they want certainty (to find/verify certain information).

KEY QUOTES

"Habits are not created, they are built upon."

"In the case of internal triggers, the information about what to do next is encoded as a learned association in the user's memory."



External → Internal triggers. Habit-forming technologies start with external triggers to prompt repeated actions, until these become closely linked to internal triggers.

- Imagine you're new to Instagram. You receive an external trigger (a friend's recommendation), which prompts you to create an account. This is followed by a series of external triggers (e.g. push notifications, friends' pictures, comments and "likes" in response to your posts), all of which prompt you to keep using Instagram. After a while, you're so used to snapping photos on Instagram that you're afraid to miss a special moment if you don't snap it.
- Social media are powerful because they have multiple internal triggers, e.g. they can be triggered by boredom, the desire to connect, fear of missing out etc.



To build the right triggers, you must first understand your users' internal triggers.

- Start with emotions (e.g. frustrations and pain-points) rather than product features. In reality, all humans are driven by a set of common needs and desires. By studying successful habit-forming products, you can understand those needs and create your own triggers.
- Examine what people actually do, rather than what they wish/say they'd do. For example, we may wish to exercise every night but end up eating junk food in front of the TV.
- To understand a behavior, keep asking "why" until you uncover the root cause or internal trigger. For example, we keep checking our emails because we're afraid to be out of the loop; we keep posting on Instagram because we're afraid to miss out on a special moment.



ACTION STEPS

Consider these questions to develop the right triggers:

- Who is the user of your product? Which internal trigger does the user experience most frequently? Express it as a statement: "Each time the user [internal trigger], he/she [first action of intended habit]."
- What's the user doing just before the intended habit? What are the best times/places to send an external trigger? How can you link an external trigger to the time the internal trigger is activated?

KEY QUOTES

- What are 3 internal triggers that can cue your user to action?
Brainstorm 3 conventional triggers using existing technology (e.g. emails, notifications) and 3 triggers that may not be viable yet (e.g. biometric sensors).



Step 2: Action



A trigger is useless unless it creates real action. And, people will only take action if the desire outweighs the effort (mental/physical).



According to Dr. BJ Fogg, Behavior = Motivation + Ability + Trigger (B = MAT). To successfully trigger action, you must make it easy to perform the action, increase the motivation to do it, *and* have an effective trigger.

- Motivation to take the action. Fogg says that there're 3 key motivators, each with a set of positive and negative feelings:
 - (i) Seek pleasure + Avoid pain
 - (ii) Seek hope + Avoid fear
 - (iii) Seek social acceptance + Avoid rejection
- Ability to complete the desired action. The fewer and simpler the steps, the more likely that people can/will do it. As a rule of thumb, always improve ability first, then motivation. Fogg presents 6 elements that affect how easy it is to complete a task:
 - (i) Time
 - (ii) Money
 - (iii) Physical effort
 - (iv) Brain cycles (i.e. the amount of mental focus and effort)
 - (v) Social deviance (how far the behavior is accepted by others)

KEY QUOTES

"To initiate action, doing must be easier than thinking."

"A given behavior will occur when motivation, ability, and a trigger are present at the same time and in sufficient degrees."

"The greatest return on investment will generally come from increasing a product's ease-of-use."

"To successfully simplify a product, we must remove obstacles that stand in the user's way."

(vi) Routine (how far it fits or disrupts existing routines).

- A trigger to activate the behavior (covered in Step 1 above).



Our decisions and behaviors are also influenced by mental biases or heuristics. For example:

- Scarcity: When something is perceived to be scarce (or to suddenly become scarce), people value it more. That's why e-commerce stores often add words like "only 7 left in stock".
- Framing: Our contexts or surroundings shape our perceptions. People pay a lot to listen to famous violinist Joshua Bell in the concert halls; yet few people even stopped to listen when Bell played in a subway station. Brain-scans also showed that people's enjoyment of wine increased when they believed it cost more (even though it was exactly the same wine).
- Anchoring: We tend to anchor our decisions to 1 piece of info, e.g. if we enter a store with a "50% off" sign, we tend to zoom in on the 50% discount rack even though there may be better items on other racks.
- Endowed progress: Customers who were given loyalty punch cards with 2 boxes punched out of 12 boxes were 82% more likely to complete the punch cards, compared with those with 0 out of 10 boxes punched. Social media sites like LinkedIn and Facebook use this heuristic to collect more data, e.g. showing progress bars to encourage people to improve their profile strength.



ACTION STEPS

- Define your users' path, from the time they experience an internal trigger till they get the desired outcome from your product. Consider how you can **remove steps** to simplify the process as much as possible.
- Which of the 6 elements above may limit the users' ability to complete the intended tasks? Brainstorm 3 testable ways to **make the tasks easier**.
- Use **heuristics** to encourage habit-formation.

KEY QUOTES



Step 3: Variable Reward



Dopamine is a pleasure-inducing chemical that's released when we experience or expect a reward. Once we learn that something predictably delivers a reward, the anticipation of the reward is already enough to activate dopamine release.

- When users' problems are solved, they're rewarded and become more motivated to repeat the action. To form habits, your product must solve a real problem consistently and reliably.
- Yet, if something becomes so predictable that we know *exactly* what will happen, we lose attention. When the cause-effect pattern is broken, we suddenly pay attention again. Variable rewards increase the likelihood that we'll keep doing something because we're anticipating a reward and also curious to see what we'll get. This increases the effects of dopamine.



Habit-forming products use 3 key types of variable rewards:

- Rewards of the Tribe. Social rewards (e.g. Facebook "likes", upvotes on forum pages, and profile badges) are driven by the human need to feel liked, accepted, connected and included. Such rewards are also variable since we can't predict others' responses (e.g. how many likes or upvotes we'd get).
- Rewards of the Hunt come from money, things or information. For example, slot machines keep you playing round after round in anticipation of hitting the jackpot. You scroll through your endless social media feeds to see what else may come up.
- Rewards of the Self tap on our desire for mastery, competence, consistency and completion, e.g. we try to level up in video games, learn a new software or win awards/accolades. Variation only makes the process more interesting.



Use these tips for designing your variable rewards:

- Use 1 or more of the 3 variable rewards above.

KEY QUOTES

"To hold our attention, products must have an ongoing degree of novelty."

- Variable rewards are only effective if the rewards match the users' internal triggers and motivations for using the product, e.g. Quora's upvoting system works because it's aligned with users' desire to find great answers.
- Give users a sense of autonomy. A habit-forming product must be enjoyable in itself and make users feel like they're in control. Give people the option to accept or refuse certain features. Tap on behaviors that users want to do and are familiar with. For example, if a health app only works when users record *every* meal and exercise, users will soon drop off. However, if the app offers virtual encouragement, comments and questions from other users in response to each new entry, people will naturally feel motivated to keep going.
- Content-driven products can be infinitely variable. However, if your product has limited or finite variability, it'll eventually become predictable unless you deliberately add new experiences to keep it novel.



ACTION STEPS

- Interview 5 customers to find out (i) what they enjoy the most about using your product, (ii) if they experience any moments of delight or surprise, and (iii) if there's anything especially satisfying about your product.
- Define the reward, i.e. how your product relieves the user's pain. Then brainstorm ways to augment it using the 3 types of variable rewards.



Step 4: Investment



In order for a behavior to become automatic, users must complete this final step—they must invest in the product by putting in their time, effort, money, social capital etc. The more users invest in your product, the more they'll return to use it because of how we rationalize our behaviors:

- Perceived value: We place an irrationally high value on our own efforts. People who made their own origami bid 5x higher prices for them compared to others who didn't. The more effort we invest in something, the more we'll value it.

KEY QUOTES

"People must want to use the service, not feel they have to."

"The more users invest time and effort into a product or service, the more they value it."

- Consistency: We tend to act consistently with past decisions. When residents were asked to place a large sign in their yard about safe driving, only 17% agreed. However, when residents were first asked to put a small sign in their window (most of whom agreed), 76% later agreed to the large sign in the yard.
- Cognitive dissonance: When there's a conflict in our thoughts, beliefs or attitudes, we'll adjust them to remove the dissonance. Even when we don't like something initially, we'll gradually shift our perception if we see others enjoying them.



Here are some tips for implementing this step effectively:

- Investment creates friction/resistance, and must happen only *after* users have already received the variable rewards.
- This investment must be associated with a potential improvement in the service or experience, e.g. recording personal preferences, inviting friends, adding content (e.g. playlist, notes, photos), building a base of followers, or learning to use an app/software.
- The best habit-forming technologies tap on users' behaviors to initiate new triggers, e.g. having users generate to-do lists that trigger new reminders, generating a list of recommendations that users can shortlist, notifying users of responses to their posts etc.



ACTION STEPS

- Review the process for using your product. Brainstorm 3 ways to **add a small user investment**, e.g. add data, content, followers, reputation, skill.
- Brainstorm how to **load the next trigger**, and how to reduce the cycle-time through the Hook.

KEY QUOTES



The 4 Steps of the Hook Model



The Manipulation Matrix



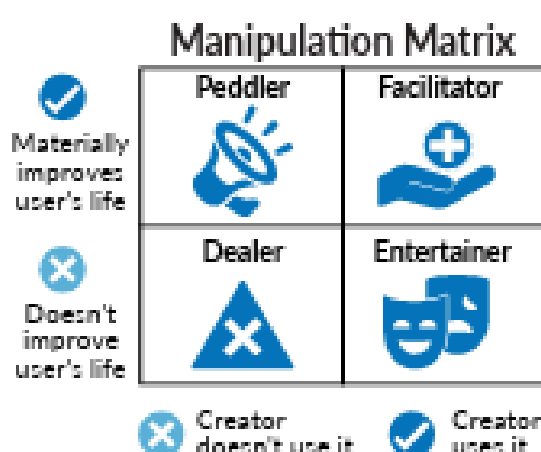
Habit-forming products are powerful since they can fundamentally change people's behaviors. And, our ability to create such products will only increase as we improve our capacity to collect and process customer data. Eyal urges innovators and businesses to use habit-forming technologies ethically and responsibly. Help people to form positive/useful habits, not negative/destructive ones.



Use the Manipulation Matrix to consider the implications of your innovation, including if you *can* and *should* hook your users.

- Ask yourself 2 questions: "Would I use this product myself?" and "Will it help users to improve their lives?"

- There are 4 key quadrants:



- (i) *Facilitators* help users to create a healthy habit. It's morally better and increases your chances of success (by delivering something of real value). However, you can't be an effective facilitator unless you've personally experienced the problem.
- (ii) *Peddlers* try to incentivize people to do things that they don't otherwise value. There's nothing morally wrong with this, but your chances of success will be low.
- (iii) *Entertainers* may not materially improve their users' lives, but they can bring temporary joy, laughter and connection. However, such rewards are short-lived, so you must constantly find new ways to inject novelty.
- (iv) *Dealers* offer a product that's exploitative, i.e. it exists mainly to extract money from the user, e.g. gambling, drugs.

KEY QUOTES

"Ubiquitous access to the web, transferring greater amounts of personal data at faster speeds than ever before, has created a more potentially addictive world."

"Peddlers tend to lack the empathy and insights needed to create something users actually want."

"Entertainment is a hits-driven business because the brain reacts to stimulus by wanting more and more of it, ever hungry for continuous novelty."



YouVersion's Bible app is a great case-study of a habit-forming product. As of Jul 2013, it has been downloaded on >100mil devices.

- The app was designed to help people with their Bible study. It started as a desktop website, but didn't really take off until they created a mobile version that could be used anytime, anywhere.
- The app was optimized for reading, with >400 reading plans to cater to a range of audience, goals, preferences and languages. It presented small, digestible chunks of daily reading (which was much less intimidating than reading the whole Bible).
- The app utilized effective triggers (e.g. greetings during Christmas, members' virtual encouragement and community emails), all of which prompted people to open the app or share the messages. User data and behavioral patterns were used to further tailor the app, to make it as easy/simple to absorb God's Word as possible. When users went through a rough patch, they often found themselves uplifted by inspiring notifications, as if they were hearing from God himself. The app also offers an element of variability and surprise, since you don't know what verse or notifications you'll get each day.
- Users progressively invest in the app when they create bookmarks, highlight verses and make notes. Over time, the app becomes an invaluable personal Bible, packed with their own insights and notes.



Habit Testing



You can use the Hook Model to identify potential weaknesses and ideas for improvement. However, you won't know which ideas will work until you test them. Eyal presents a simple Habit Testing process based on the build-measure-learn Lean method. This is a continual process that should be applied to every new product feature and iteration:

- Identify what a devoted user means. Consider who are your product's habitual users and how often they'd realistically

KEY QUOTES

"It's a case study in how technology can change behavior by marrying the principles of consumer psychology with the latest in big data analytics."

use the product. Define the numbers and type of users who meet these criteria: if <5% of your users are devoted users, it probably means your product isn't valuable or useful enough.

- Codify the Habit Path. Different customers will use your product differently. Your goal is to collect enough data to identify a pattern or a "Habit Path" shared by your most loyal users. Then, seek to replicate those action steps. For example, Twitter found (in its early days) that once new users followed 30 other members, they were drastically more likely to keep using the app.
- Modify or gently prod new users down the Habit Path.



Uncover Habit-forming Opportunities



You can discover great opportunities through careful observation:

- Daily behaviors. Consider why you do or don't do certain things, and how those tasks could be made easier and/or more rewarding. Ask yourself: "what problem do I wish someone else could solve for me?"
- Nascent behaviors. Technologies that are massively successful in a niche often end up spreading if they can address a wider need. Such innovations (e.g. disposable cameras, telephones) are often dismissed initially as nice-to-have vitamins but eventually become a painkiller for a major pain/itch.
- Enabling technologies. Every 10 years or so, new enabling technologies (e.g. the internet) emerge to provide the foundation for mass market penetration/adoption. Alternatively, a shift in user interface (i.e. how people interact with technology) can also bring massive changes, like how Apple changed the mobile phone interface and Google changed the search interface.

KEY QUOTES

"Careful introspection can uncover opportunities for building habit-forming products."

"Wherever new technologies suddenly make a behavior easier, new possibilities are born."

ACTION STEPS

- Consider where your product falls on the Manipulation Matrix and if you're proud of the way you're influencing people.
- Use **habit testing** to identify the Habit Path for long-term engagement.
- Look for new **habit-forming opportunities** by observing your own behaviors, looking for nascent behaviors, enabling tech or new interfaces.



Other Details in the Book to Look out For

This is an easy-to-read book that includes many real-life examples, case studies and useful empirical research/background. Each chapter comes with a brief summary and recommended questions/exercises to help you get started. We've captured the key ideas and tips with selected examples in this summary. For more details, please visit www.nirandfar.com/hooked/.

About the Author

Nir Eyal is an Israeli-born American author, lecturer and investor. He earned a B.A. at Emory University, worked for Boston Consulting Group and a solar panel installation firm before graduating with a Master of Business Administration from Stanford. He then founded a digital marketing company with his friends before becoming a consultant in product design. Eyal also teaches university courses and gives speeches about how psychology intersects with technology and business.



KEY QUOTES